

Smart Traffic Lights using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded. Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. This paper presents the concept of using local AI processing for smart traffic-light control, with an additional focus on emergency vehicle priority.

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded. Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. For beginners, the main idea is simple: instead of changing lights only according to a clock, the system uses information from the road. Research reviews also show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

II. CLOUD-BASED AND LOCAL AI SYSTEMS

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay. Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions. A small local computer can examine camera input and send a short instruction to the traffic-light controller.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

III. EMERGENCY VEHICLE PRIORITY

Emergency vehicles such as ambulances, fire trucks, and police vehicles may need traffic signals to provide them with priority. A study by Asaduzzaman and Vidyasankar proposed a priority algorithm for controlling traffic signals for emergency vehicles. The study discusses Emergency Vehicle Preemption (EVP) and uses signal-priority techniques to serve emergency vehicles while also considering the effect on general traffic [3].

This provides a research basis for extending a local smart-camera traffic system so that it can detect an emergency vehicle and coordinate nearby traffic signals to create a clearer route.

IV. SYSTEM COMPARISON

The systems can be compared using three simple measures. Basic Timed Lights make decisions every 30 seconds, use zero internet data, and have a setup cost of \$200. Cloud AI Systems make decisions in 2 seconds, use high internet data, and have a setup cost of \$2,500. Local Smart Cameras make decisions in 0.5 seconds, use low internet data, and have a setup cost of \$1,200.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

V. SIMPLE SYSTEM ARCHITECTURE

The proposed architecture follows four steps. First, Street Cameras capture traffic conditions. Second, the video information is sent to Smart Pole Computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the City Cloud Server for storage and wider analysis. Fourth, the decision is sent to the Traffic Lights, which change their signals.

The flow is therefore:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

VI. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance. Research on emergency vehicle preemption has explored priority-based traffic signal control to allow emergency vehicles to pass through intersections more efficiently [3].

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

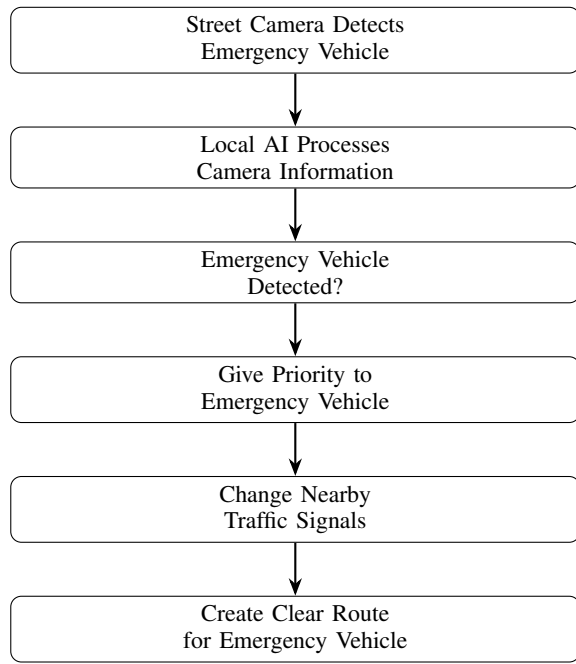


Fig. 1. Proposed Local AI-Based Emergency Vehicle Priority Flow

VII. CONCLUSION

Smart traffic lights using local AI can use camera information to respond to changing traffic conditions while reducing dependence on continuous cloud processing. Machine-vision methods can support adaptive traffic-light control, while emergency vehicle priority algorithms can provide a basis for creating clearer routes for ambulances and other emergency vehicles. Combining these ideas can form a useful direction for future smart traffic systems.

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